

1-Year Roadmap: Data Analytics, Data Science & GenAI

Goal: Placement-Ready for roles like Data Analyst, Data Scientist, GenAI Engineer, LLM App Developer

Module 0: Orientation & Core Concepts

Duration: 1-2 Days

Goal: Get students familiar with industry terms, career paths, tools, and how everything connects.

Basic Industry Terms to Understand

- What is Data (Structured, Semi-Structured, Unstructured)
- Difference between Data Analytics, Data Science, Machine Learning, and Artificial Intelligence
- What is Generative AI (ChatGPT, DALL·E, etc.)
- What is Cloud Computing and why it matters in Data/AI
- What are Databases (SQL vs NoSQL)

Career Roles You Can Target

- Data Analyst
- BI Analyst
- Data Scientist
- ML Engineer
- Gen AI Engineer
- Prompt Engineer
- Data Engineer
- Cloud Data Engineer
- AI Product Analyst

Tools You'll Learn Throughout the Course

Category	Tools You'll Use
Programming	Python, SQL
Analysis & BI	Excel, Power BI
Data Science & ML	Pandas, NumPy, Scikit-learn, XGBoost
GenAI & Prompting	ChatGPT, OpenAI API, LangChain, Hugging face, Streamlit, Agno
Databases	MySQL, PostgreSQL
Cloud & Deployment	AWS (S3, EC2, RDS), Azure basics, bedrock, Docker
Version Control	Git & GitHub

Module 1: Python Programming for Data & AI (4 weeks)

Duration: Month 1

Topics:

- Python basics: variables, loops, conditions, functions
- Data structures: list, dict, set, tuple
- File handling, JSON, APIs
- OOPs basics
- Exception handling

Tools:

- Python (Jupyter/Colab)
- ChatGPT for code explanation/debugging

Mini Project:

- Basic data scraper & analyzer
- Python Quiz App

Mock Interview: Python + Logic + File Handling

Module 2: Data Handling with Pandas & Numpy (4 weeks)

Duration: Month 2

Topics:

- NumPy: arrays, indexing, broadcasting
- Pandas: DataFrames, Series, filtering, groupby
- Missing data handling
- Data wrangling (reshape, merge, pivot)

Tools:

- Pandas, NumPy, ChatGPT, Google Colab

Mini Project:

- IPL or Netflix dataset analysis using Pandas

Mock Interview: Data Manipulation + Real-world Datasets

Module 3: SQL & Excel for Analysts (3 weeks)

Duration: Month 3

Topics:

- SQL: SELECT, JOINS, GROUP BY, Subqueries, Window Functions
- Excel: formulas, pivot tables, charts, lookup, dashboards
- SQL for case studies

Tools:

- MySQL/PostgreSQL (pgAdmin), Excel, Formula Bot

Mini Project:

- Employee or Sales DB case study
- Excel Dashboard (e.g. Superstore or Ecommerce)

Mock Interview: SQL Joins + Case Study Discussion

Module 4: Data Visualization & Storytelling (3 weeks)

Duration: Month 4

Topics:

- Matplotlib, Seaborn, Plotly
- Creating interactive dashboards
- Power BI or Tableau (choose one)
- Storytelling with data

Tools:

- Power BI

Mini Project:

- Power BI Dashboard (e.g. COVID, Sales, HR analytics)

Mock Interview: Visualization Tools + Data Storytelling

Module 5: Statistics & Probability for Data Science (4 weeks)

Duration: Month 5

Topics:

- Descriptive statistics
- Probability distributions
- Sampling, CLT
- Hypothesis testing, t-tests, chi-square
- Correlation & Regression

Tools:

- Python (Scipy, Statsmodels), ChatGPT for formulas

Mini Project:

- A/B Testing Simulation
- Customer Churn Statistical Analysis

Mock Interview: Stats Scenarios + Probabilities + Tests

Module 6: Machine Learning - I (Supervised Learning) (4 weeks)

Duration: Month 6

Topics:

- ML pipeline
- Regression: Linear, Logistic
- Decision Tree, Random Forest, KNN
- Train-test split, model evaluation

Tools:

- Scikit-learn, Google Colab, ChatGPT, PyCaret (optional)

Mini Project:

- Loan Approval or House Price Prediction
- Predict Diabetes from health dataset

Mock Interview: Supervised Learning Models + Metrics

Module 7: Machine Learning - II (Unsupervised & Feature Engineering) (3 weeks)

Duration: Month 7

Topics:

- KMeans Clustering
- Dimensionality Reduction: PCA
- Feature selection, encoding, scaling
- Model tuning (GridSearchCV)

Tools:

- Scikit-learn, Seaborn, Colab

Mini Project:

- Customer Segmentation
- E-commerce Product Clustering

Mock Interview: Unsupervised Learning + Features**Module 8: Time Series + Intro to Deep Learning (3 weeks)****Duration: Month 8****Topics:**

- Time Series components, lag, rolling stats
- Forecasting: ARIMA, Prophet
- Intro to Neural Networks
- Perceptron, basic Keras model

Tools:

- Statsmodels, Prophet, Keras, TensorFlow

Mini Project:

- Stock Price or Weather Forecasting

Mock Interview: Forecasting + Neural Net Basics**Module 8.5: Big Data with PySpark (2 weeks)****Duration: Between Month 8-9****Topics:**

- What is Big Data?
- PySpark architecture & setup
- DataFrames, RDDs, Transformations, Actions
- PySpark SQL + basic MLlib
- Databricks (optional)

Tools:

- PySpark, Google Colab (Findspark), Databricks

Mini Project:

- Analyze NYC Taxi Dataset using PySpark

Mock Interview: PySpark Concepts + Big Data Thinking**Module 9: Natural Language Processing (NLP) (4 weeks)****Duration: Month 9****Topics:**

- Text cleaning, tokenization
- TF-IDF, word embeddings
- Sentiment analysis
- Text classification (Naive Bayes, Logistic)

Tools:

- NLTK, SpaCy, Scikit-learn

Mini Project:

- Sentiment analysis on Twitter or Reviews
- Resume Classifier

Mock Interview: NLP Use Cases + Classification**Module 10: GenAI Foundations (4 weeks)****Duration: Month 10****Topics:**

- What is GenAI & LLMs
- Prompt engineering
- Embeddings, Vector DBs (FAISS)
- OpenAI API, Anthropic, Gemini

Tools:

- OpenAI, ChatGPT, Claude, Gemini, Replit, Colab

Mini Project:

- Build a chatbot using OpenAI
- Auto-summarizer using ChatGPT API

Mock Interview: Prompting, GenAI APIs, RAG Concept**Module 11: GenAI App Building (4 weeks)****Duration: Month 11****Topics:**

- LangChain basics
- LlamaIndex, RAG pipeline
- Build workflows with Flowise
- Deploy with Streamlit/Gradio

Tools:

- LangChain, Streamlit, Hugging Face, Vercel

Mini Project:

- Resume Q&A bot using LangChain
- Voice-to-Summary App using Whisper + GPT

Mock Interview: LLM App Architecture + RAG Debugging**Module 12: GenAI Tools Lab & AI Playground****Duration: Month 11**

Goal: Hands-on practice with advanced AI tools using Python – build, integrate, and deploy mini AI solutions.

Key Focus Areas**AI Tools with Python:**

- OpenAI API (ChatGPT, Whisper, DALL·E)
- Hugging Face (Transformers, CLI)
- LangChain (for chatbots & automation)

- PandasAI (dataframe + GPT)
- AutoGPT / BabyAGI (autonomous agents)
- Replicate (ML models via API)
- LlamaIndex (custom document Q&A)
- Weights & Biases (experiment tracking)

Mini Project Lab:

- Resume screener, report generator, image-to-text, etc.

Cloud Tools:

- Google Colab, Hugging Face Spaces, Streamlit Cloud

Capstone Activity:

- Build & demo your own GenAI tool using 2+ APIs

Mock Interview (Tool-Based)

- Questions + project walkthrough focused on AI tools

Module 13: Portfolio + Mock Interviews + Job Prep (Final Month)

Duration: Month 12

Topics:

- Resume building (AI optimized)
- Mock interviews (DA, DS, GenAI)
- LinkedIn & GitHub projects
- Capstone project (group/solo)
- Freelancing & internships guide

Capstone Projects (Pick 1-2):

- Data Analytics Dashboard (end-to-end BI project)
- ML Web App (e.g., Predict & display output with Streamlit)
- LLM Chatbot with LangChain + Vector DB
- YouTube Transcript Analyzer using Whisper + GPT

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